Extra Activity - 3

The due date for submitting this assignment has passed.
Due on 2026-04-17, 23:59 IST.
You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-04-08, 08:38 IST

In this activity, we will go through a problem which demonstrates how the projection operator can be used to give a quantitative estimate to measure how good a straight line is, for modeling given data points. Let us consider two quantities xx and yy, whose dependence on each other we would like to understand. To do so, we perform experiments and observe finitely many data points. We would like to determine the straight line passing through the origin that best models the given data.

(Note that a straight line may not be the best curve to model the given data.This activity only focuses on finding the best model among straight lines passing through the origin.)

Consider the following data comprising tuples (x, y)(x, y).

v_1 v1 = (−2, −2), v_2 v2 = (−1, −1), v_3 v3 = (0, 1), v_4 v4 = (3, 2).

Answer the following questions to formulate the function that needs to be optimized in order to find the 'best straight line passing through the origin' to model the given data. In all the questions that follow, l_m lm will be used to denote the straight line in R^2 R2 of slope m that passes through the origin. $P_W(v)$ PW (v) will be used to denote the projection of the vector vv on to the subspace WW.

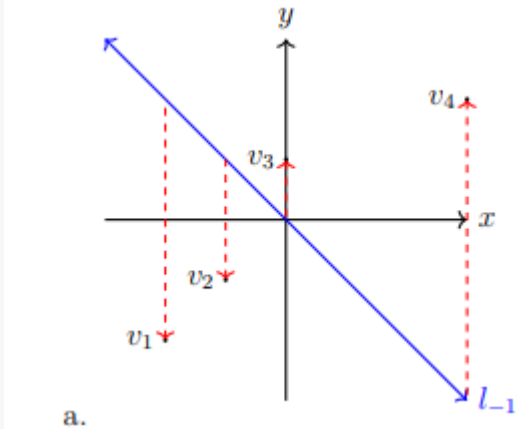
What is the maximum number of points from the above collection that can lie on a single line through the origin in R^2 R2 ?
2

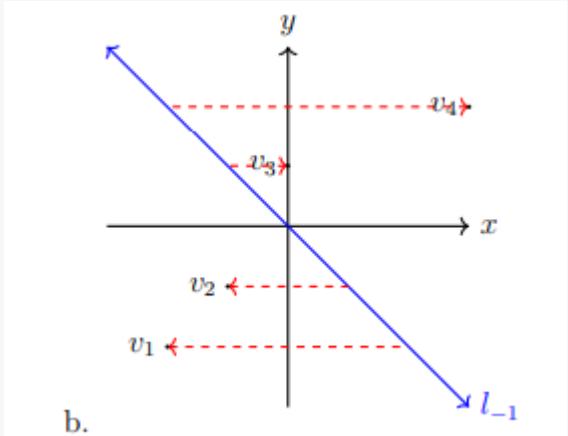
Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 2
1 point

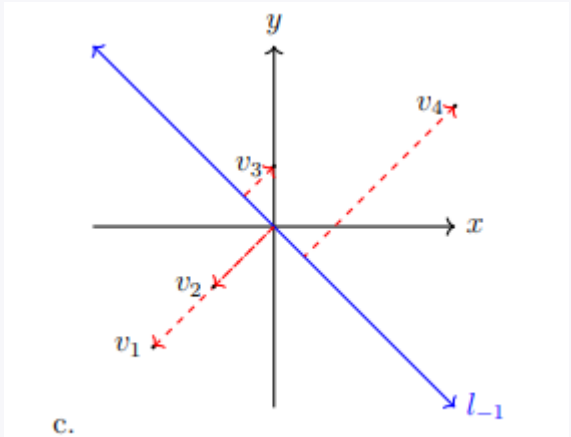
There is no single line which contains all of the given points. If l_m lm is chosen as a model for the given data, we measure the error contributed by the point v_i vifor this model using the square of the distance between v_i vi and the point on l_m lm that is nearest to v_i vi. This can be calculated using the projection operator with respect to the standard inner product. Calculate the error contributed by v_2 v2 if l_{-1} l−1 is chosen to model the given data.
2

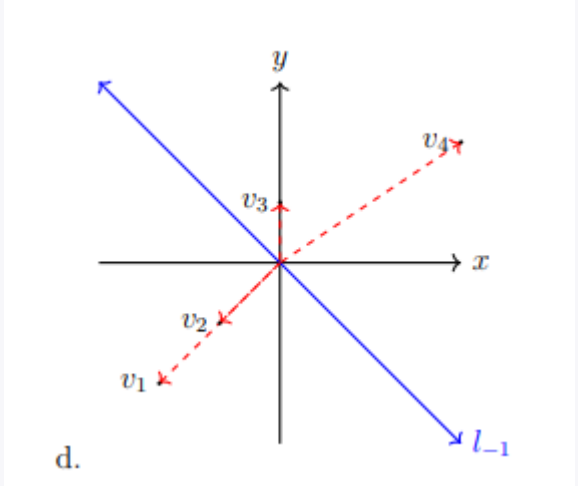
Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 2
1 point
1 point

The vector to v_i vifrom the point on the model line that is nearest to v_i viis called the error vector corresponding to v_i vi. Which of the following diagrams correctly represents the error vectors (denoted using red dashed lines)?



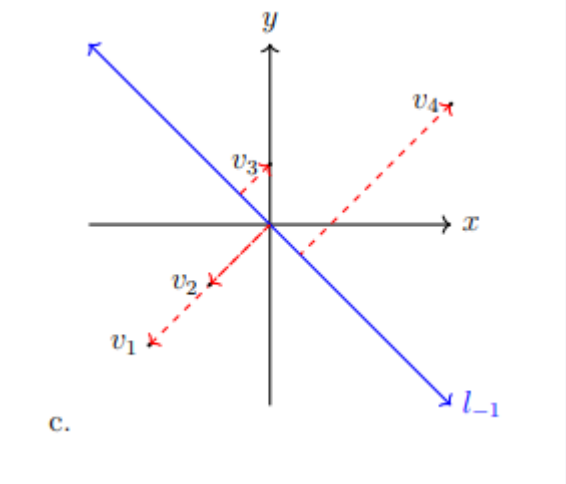






Yes, the answer is correct.
Score: 1

Accepted Answers:



1 point

The error function corresponding to a given line is defined as the sum of the errors contributed by each point. Which of the following is the correct formula for the error function for the given points modeled using the line l_m lm?

- $\sum_{i=1}^{4} ||v_i||^2$ ∑i=14 ||vi||2
- $\sum_{i=1}^{4} ||P_{l_m}(v_i)||^2$ ∑i=14 ||Plm(vi)||2
- $\sum_{i=1}^{4} ||v_i + P_{l_m}(v_i)||^2$ ∑i=14 ||vi + Plm(vi)||2
- $\sum_{i=1}^{4} ||v_i - P_{l_m}(v_i)||^2$ ∑i=14 ||vi − Plm(vi)||2

Yes, the answer is correct.

Score: 1

Accepted Answers:

$\sum_{i=1}^{4} ||v_i - P_{l_m}(v_i)||^2$ ∑i=14 ||vi − Plm(vi)||2

Techniques from calculus can be used to find the minimum of the above function. This gives the value of the slope for which the error is minimum. This line is called the line of best fit. You are encouraged to